Advanced Topics in Databases: Presentation

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Introduction

Goal: To design and implement a database system to manage and analyze election data, including the development of an ETL pipeline, functions, views, triggers, and a frontend for data interaction.

Architecture

- The architecture of our system consists of three main components: the database, the ETL pipeline, and the frontend.
- The database is designed to efficiently store and manage the election data, while the ETL pipeline is responsible for extracting, transforming, and loading the data into the database.
- The frontend provides a user-friendly interface for users to interact with the data and visualize the results.

Database Design

- We designed a database schema to efficiently store the election data, which included tables for elections, municipalities, parties, candidacies, votes, and seat allocations.
- The schema was designed to allow for efficient querying and analysis of the election data, while also ensuring data integrity and consistency.

ETL Pipeline

- We designed and implemented a robust ETL pipeline to handle the election data, which involved extracting data from various sources, transforming it to fit our database schema, and loading it into the database efficiently.
- The ETL pipeline was designed to be scalable and flexible, allowing us to handle multiple elections and municipalities without significant changes to the code.

Analytical Queries

- We implemented various functions, views, and triggers to perform necessary calculations and maintain data integrity.
- We tested these implementations using SQL queries to ensure they were working correctly and efficiently.
- We analyzed the results to gain insights into the election data, such as identifying which parties had the most votes and how the seats were allocated based on the D'Hondt method.

Frontend Overview

- We developed a frontend using Flask to create a web application that allows users to interact with the election data.
- The frontend provides a user-friendly interface to query the database and visualize the results, such as displaying vote percentages and seat allocations in a more intuitive way.
- It also allows users to filter results by election, municipality, and party, making it easier to analyze specific aspects of the election data.
- Overall, the frontend enhances the accessibility and usability of the election data, allowing users to explore and analyze it in a more interactive way.

Conclusion

We managed to design and implement a comprehensive database system to manage and analyze election data, including a robust ETL pipeline, various functions, views, triggers, and a user-friendly frontend. Through this project, we gained valuable insights into the election data and were able to analyze it in various ways, such as identifying which parties had the most votes and how the seats were allocated based on the D'Hondt method. Overall, this project was a great learning experience that allowed us to deepen our understanding of databases and their applications in real-world scenarios.